

# SPARC II

Shuffled-Radius Nulls, Halo-Catalogue Cross-Check Interface, Morphology Splits, and an External  $Q_{\text{bar}}$  Check

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## Abstract

Paper 61 introduced a first SPARC real-data pilot for MAAT structural dark-residual diagnostics. It showed that the structural diagnostic  $D_{\text{struct}}$  carries non-random galaxy-level signal relative to lightweight NFW-like and RAR/MOND-like comparison channels. This paper performs the predeclared robustness follow-up.

We test four upgrades. First, a shuffled-radius null destroys the radial pairing of  $D_{\text{struct}}(r)$  within each galaxy while preserving each galaxy’s distribution of structural scores. Second, galaxies are split into baryonic morphology proxies derived from the available SPARC rotation-curve mass components. Third, the code implements an optional published halo-catalogue cross-check interface for Li-et-al.-style SPARC halo-fit tables. No such local halo-catalogue table is redistributed in the present repository, so the catalogue cross-check is reported as not run in this execution. Fourth, we add a neutral external comparison against a supplied HGD–GSR galaxy-level baryonic concentration proxy

$$Q_{\text{bar}} = \frac{\text{median}(V_{\text{bar,inner}})}{\text{median}(V_{\text{bar,outer}})}.$$

This table covers the full 175-galaxy SPARC sample and is treated as an external input rather than as a MAAT-derived quantity.

The radius-null result is channel-dependent. The primary screening statistic is the pooled within-galaxy radial correlation between  $D_{\text{struct}}$  and the internal NFW-like residual proxy,

$$\rho_S = 0.3930,$$

far above the shuffled-radius 95% absolute null threshold

$$|\rho_S|_{95} = 0.0355.$$

By contrast, the RAR channel gives only

$$\rho_S = 0.0123,$$

below its shuffled-radius 95% threshold. Thus the Paper-61 signal survives the stronger radial null in the NFW-like proxy channel but not in the RAR channel. Because the NFW-like channel is only an internal lightweight proxy, this result is best read as a hypothesis-generating robustness signal, not as a validation against professional halo modelling.

Morphology-proxy splits show that the mean structural residual is largest in dwarf-like and dark-matter-dominated disk proxies, while the strongest radial NFW-like alignment appears in bulge and ordinary disk proxies. The external  $Q_{\text{bar}}$  check finds that  $Q_{\text{bar}}$  is strongly associated with rotation speed and bulge proxy structure, and is anti-correlated with  $D_{\text{struct}}$ :

$$\rho_S(Q_{\text{bar}}, \langle D_{\text{struct}} \rangle) = -0.4345.$$

After controlling for simple rotation-speed and baryonic proxy variables, an attenuated but nonzero partial rank association remains. The result is therefore not a universal dark-matter claim. It is a more precise statement: the MAAT structural residual carries radius-resolved SPARC information in some channels and galaxy populations, the external  $Q_{\text{bar}}$  signal is partly complementary but strongly morphology-linked, and the full published-halo-catalogue comparison remains future work until the external Li-et-al. table is supplied locally.

## 1 Purpose

Paper 61 was deliberately modest. It did not claim that MAAT explains dark matter, replaces cold dark matter, or outperforms professional halo modelling. It asked a smaller question:

*Can the structural residual diagnostic be applied to real SPARC rotation curves, and does it carry non-random signal?*

Paper 65 asks a stricter question. If  $D_{\text{struct}}$  is meaningful, it should not merely survive galaxy-level shuffling. It should also retain signal when the radial ordering inside each galaxy is tested. Moreover, the signal should be inspected across galaxy populations and prepared for comparison with published SPARC halo-fit catalogues.

The interpretive hierarchy is therefore deliberately conservative. The strongest possible version of this paper would require a machine-readable published halo-fit catalogue, official morphology information, and a multivariate treatment of baryonic concentration proxies. The present paper does not yet have all three. It should be read as a robustness and signal-localisation step:

- the NFW-like channel is a lightweight shape proxy, not a professional halo fit;
- the RAR channel is an important negative control and fails the radial null in this run;
- the morphology classes are operational proxy splits, not official Hubble types;
- the external  $Q_{\text{bar}}$  check is exploratory and post-hoc;
- $D_{\text{struct}}$  is imported from the Paper-61 diagnostic pipeline rather than rederived from first principles here.

Accordingly, the numerical effect size in the NFW-like channel should not be read as a standalone astrophysical claim. Its main value is to identify a non-random channel that deserves a stricter confirmatory test against published halo catalogues.

## 2 Inputs

The input data are the Paper-61 SPARC-derived radius diagnostics:

`paper61_radius_diagnostics.csv`,      `paper61_galaxy_summary.csv`.

These are derived from the SPARC rotation-curve data of Lelli, McGaugh, and Schombert [1]. The present run contains

$$N_{\text{gal}} = 175, \quad N_{\text{rad}} = 3391.$$

The SPARC input is used with the same fixed stellar mass-to-light factors as Paper 61,

$$\Upsilon_{\text{disk}} = 0.5, \quad \Upsilon_{\text{bulge}} = 0.7.$$

This is still a simplifying assumption. Paper 65 is a robustness diagnostic, not a precision galaxy-fit analysis.

**Operational definition of  $D_{\text{struct}}$ .** The structural diagnostic  $D_{\text{struct}}$  is not newly derived in this paper. It is inherited from the Paper-61 MAAT/SPARC pipeline as a radius-level diagnostic constructed from baryonic, residual, and robustness channels. For transparency, the code-level construction is:

$$V_{\text{bar}}^2 = V_{\text{gas}}|V_{\text{gas}}| + \Upsilon_{\text{disk}}V_{\text{disk}}|V_{\text{disk}}| + \Upsilon_{\text{bulge}}V_{\text{bulge}}|V_{\text{bulge}}|,$$

$$f_{\text{res}}(r) = \frac{\max(V_{\text{obs}}^2 - V_{\text{bar}}^2, 0)}{\max(V_{\text{obs}}^2, \epsilon)}, \quad R_{\text{rob}} = \min\left[(HBV)^{1/3}, (HBS_{\text{eff}}V)^{1/4}\right],$$

$$D_{\text{struct}}(r) = f_{\text{res}}(r) R_{\text{rob}}(r) V(r).$$

Here  $H$  downweights baryonic overshoot,  $B$  downweights steep or strongly curved rotation-curve structure,  $S_{\text{eff}}$  gates residual-gradient activity, and  $V$  measures radial overlap and shape compatibility between residual-like and baryonic density proxies. Thus  $D_{\text{struct}}$  is not a fitted halo parameter. It is an operational structural-residual score: large where the residual fraction is nonzero and passes the local robustness and radial-compatibility gates. This paper asks whether the radial placement of that already defined diagnostic carries non-random information under stricter null tests. Consequently, the present paper is not a standalone theoretical justification of  $D_{\text{struct}}$ . It is an empirical stress test of that diagnostic on SPARC-derived data.

### 3 Shuffled-Radius Null

Paper 61 used a shuffled-galaxy null: galaxy-level values of  $D_{\text{struct}}$  were permuted against mismatch channels. Paper 65 adds a stronger radial null. For each galaxy  $g$ , the observed set

$$\{D_{\text{struct}}(r_i)\}_{i \in g}$$

is preserved, but randomly reassigned to the radii of the same galaxy. This preserves:

- the galaxy identity;
- the distribution of  $D_{\text{struct}}$  values in that galaxy;
- the number and spacing of radial samples;
- the galaxy-level residual amplitude.

It destroys only the radial pairing between  $D_{\text{struct}}(r)$  and the comparison channels.

Before pooling, each quantity is standardised within each galaxy. The reported statistic is therefore a within-galaxy pooled Spearman correlation:

$$\rho_S(z_g[D_{\text{struct}}(r)], z_g[M(r)]),$$

where  $M(r)$  is either the NFW-like fractional residual channel or the RAR/MOND-like sigma residual channel.

### 4 Halo-Catalogue Cross-Check Interface

Li et al. [6] provide a comprehensive catalogue of dark-matter halo models for SPARC galaxies, fitting seven halo profiles including pISO, Burkert, NFW, Einasto, DC14, coreNFW, and Lucky13. Those published fits are the correct next external baseline.

The Paper-65 code therefore accepts an optional local CSV file:

```
python3 sparc_ii_paper65.py --halo-catalog path/to/li_halo_catalog.csv
```

The expected minimal columns are:

`galaxy,      chi2`

with flexible aliases for names and reduced- $\chi^2$ -like columns. For multi-model catalogues, the script keeps the best available fit per galaxy and reports the correlation between  $D_{\text{struct}}$  and the published halo-fit mismatch.

**Status in this run.** No local Li-et-al. halo-catalogue table is present in the repository. Therefore the published catalogue comparison is not executed here. The figure and JSON output explicitly mark this as

`not_run_no_local_catalogue.`

The internal NFW-like channel remains a lightweight proxy, not a professional halo-fit substitute. This point constrains the interpretation of the positive NFW-like result. Passing the shuffled-radius null against this channel shows that the structural diagnostic tracks non-random radial information relative to the internal NFW-like mismatch proxy. It does not show that  $D_{\text{struct}}$  outperforms, replaces, or explains published NFW-family halo fits. Until the Li-et-al. catalogue comparison is executed, the positive NFW-like result remains a screening result rather than a model-comparison result.

## 5 Morphology and Proxy Splits

The prepared Paper-61 rotation-curve CSV does not include Hubble type. The code therefore supports two modes:

- if a local morphology catalogue is supplied, it is joined by galaxy name;
- otherwise, baryonic morphology proxies are derived from the available SPARC mass components.

The fallback proxy classes are:

- `bulge_proxy;`
- `disk_proxy;`
- `dwarf_proxy;`
- `gas_rich_late_proxy;`
- `dm_dominated_disk_proxy.`

These are not official Hubble types. They are operational population splits based on maximum rotation speed, gas fraction proxy, bulge fraction proxy, and outer residual fraction. They are useful for detecting whether a signal is concentrated in obvious galaxy populations, but they should not be interpreted as a clean astrophysical morphology catalogue. For this reason, the morphology-proxy split is descriptive only and is the weakest population-level part of the present analysis. It can reveal whether an effect is plausibly population-dependent, but it cannot by itself establish a robust morphology law, identify Hubble-type dependence, or support parameter-level inference about galaxy classes.

## 6 Results

### 6.1 Radius-Shuffle Null

| Channel           | Real $\rho_S$ | Null $ \rho_S _{95}$ | Exceeds? |
|-------------------|---------------|----------------------|----------|
| NFW-like residual | 0.3930        | 0.0355               | yes      |
| RAR residual      | 0.0123        | 0.0336               | no       |
| Radius itself     | 0.1241        | 0.0332               | yes      |

Table 1: Within-galaxy pooled shuffled-radius null results.

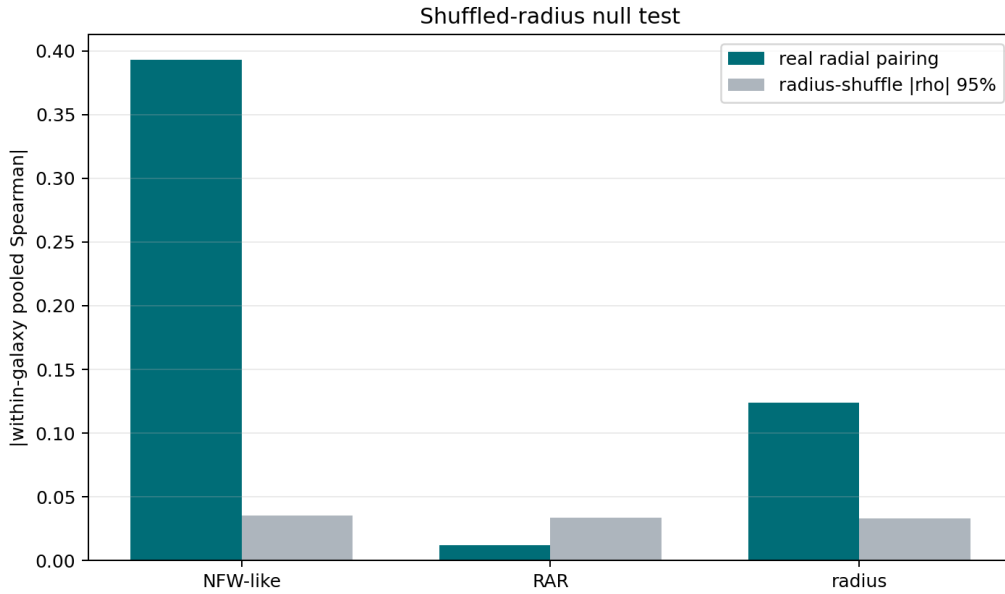


Figure 1: Shuffled-radius null test. The NFW-like channel remains far above the radius-shuffled null, while the RAR channel does not.

This is the main Paper-65 screening result. The structural diagnostic is not merely a galaxy-level pairing artifact in the NFW-like proxy channel: its radial ordering carries information relative to that proxy. However, this is a technical robustness result, not a physical conclusion about dark-matter halo models. The result is also not channel-universal. The RAR channel does not survive the same radial test. This failure is not a side issue. It is one of the main constraints on the claim. It implies that  $D_{\text{struct}}$  is not simply detecting every reasonable rotation-curve comparison channel. Rather, it appears to align with the internal NFW-like mismatch proxy while remaining largely absent in the RAR/MOND-like residual channel under the same radial null. There are at least three conservative readings. First, the RAR comparison may already absorb much of the baryonic radial information to which  $D_{\text{struct}}$  is sensitive, making the remaining  $D_{\text{struct}}$ -RAR residual relation weak or redundant. Second, the present RAR channel is a lightweight sigma-residual implementation, not a full hierarchical RAR fit with galaxy-specific scatter and uncertainty modelling. Third,  $D_{\text{struct}}$  may be more sensitive to residual-shape structure than to acceleration-relation deviations. The present data do not choose between these explanations. Any future physical interpretation must explain this channel asymmetry rather than treating the NFW-like positive result in isolation.

## 6.2 Galaxywise Radial Correlations

For galaxies with at least five radial samples, the median galaxywise correlations are:

| Quantity                                 | Median galaxywise $\rho_S$ |
|------------------------------------------|----------------------------|
| $D_{\text{struct}}$ vs NFW-like residual | 0.3000                     |
| $D_{\text{struct}}$ vs RAR residual      | -0.1227                    |
| $D_{\text{struct}}$ vs radius            | -0.0350                    |

Table 2: Galaxywise radial correlation medians for 171 galaxies with at least five radial points.

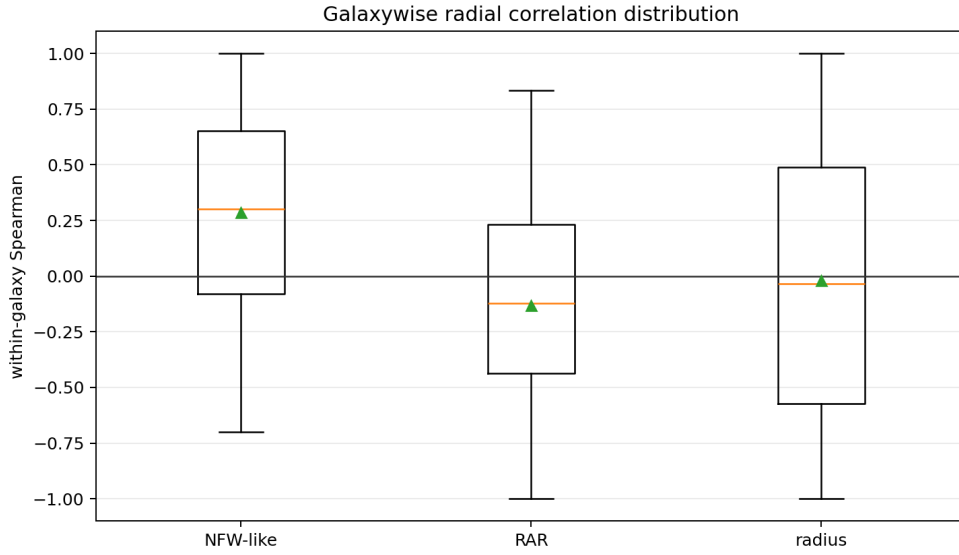


Figure 2: Distribution of galaxywise radial correlations.

## 6.3 Morphology-Proxy Splits

| Proxy class                      | Galaxies | $\langle D_{\text{struct}} \rangle$ | NFW radial $\rho_S$ | RAR radial $\rho_S$ |
|----------------------------------|----------|-------------------------------------|---------------------|---------------------|
| bulge proxy                      | 26       | 0.0883                              | 0.6473              | 0.2070              |
| disk proxy                       | 38       | 0.1218                              | 0.4536              | -0.0668             |
| dark-matter-dominated disk proxy | 32       | 0.2191                              | -0.0104             | 0.1564              |
| dwarf proxy                      | 70       | 0.2436                              | 0.2402              | -0.2372             |
| gas-rich late proxy              | 9        | 0.2193                              | 0.2394              | 0.1416              |

Table 3: Baryonic morphology-proxy splits. These are not official Hubble types; they are operational classes derived from the available mass-model components and should be read only as descriptive sanity checks.

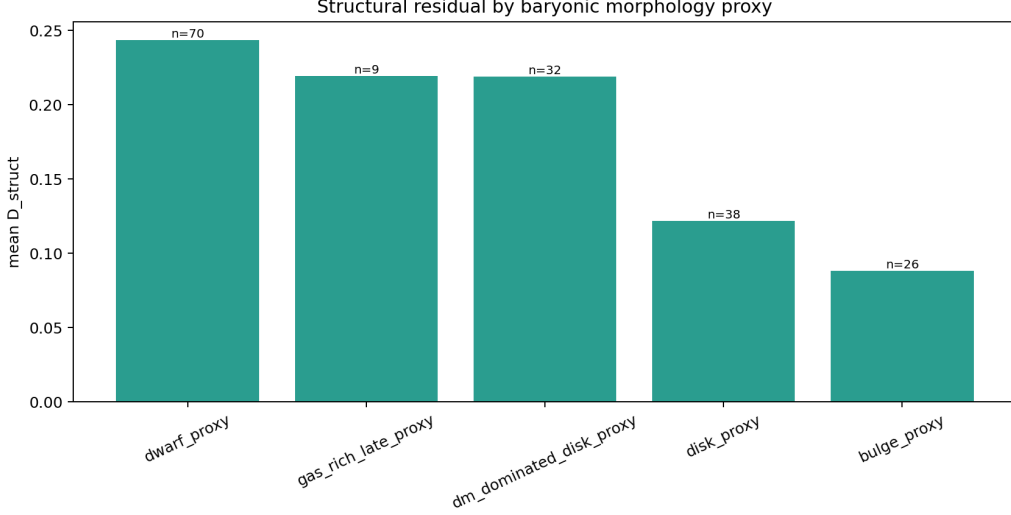


Figure 3: Mean structural residual by baryonic morphology proxy. Dwarf-like and dark-matter-dominated disk proxies have the largest average structural residual, while bulge/disk proxies show stronger radial NFW-like alignment. Because these are proxy classes, the plot is descriptive rather than a morphological inference.

#### 6.4 Halo-Catalogue Slot

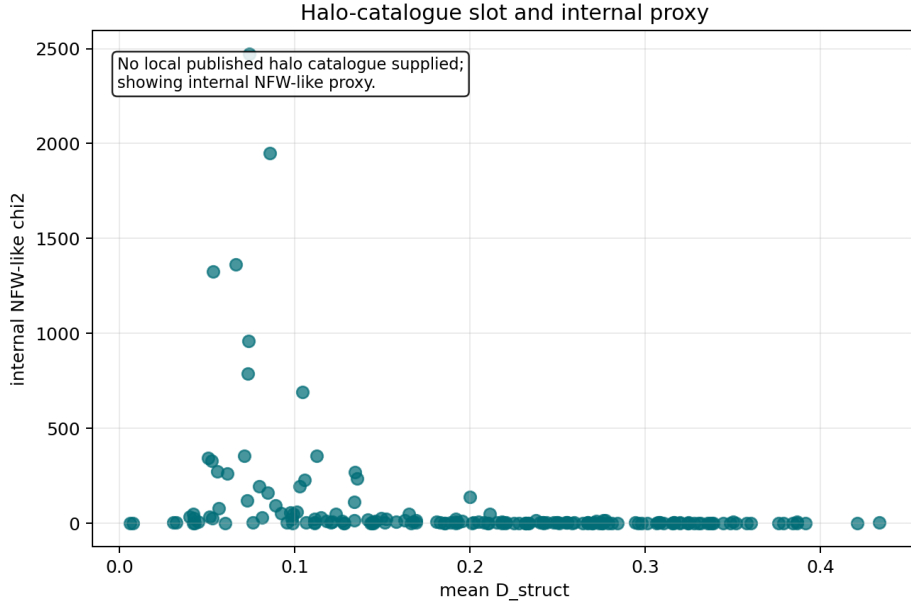


Figure 4: Published halo-catalogue cross-check slot. Since no local Li-et-al. table is supplied in this run, the plot shows the internal NFW-like proxy and marks the catalogue comparison as not yet executed.

#### 6.5 Exploratory Post-Hoc $Q_{\text{bar}}$ Cross-Check

After the Paper-65 run, an external galaxy-level baryonic concentration proxy was supplied in the form

$$Q_{\text{bar}} = \frac{\text{median}(V_{\text{bar,inner}})}{\text{median}(V_{\text{bar,outer}})}.$$

The inner and outer regions are defined as the first and last 20% of radial points, respectively. The table contains one  $Q_{\text{bar}}$  value for each of the 175 SPARC galaxies used in the present analysis. We treat this table as a neutral external input. It is not used to define  $D_{\text{struct}}$ , and the Paper-61/Paper-65 pipeline is not refit or modified. Because this comparison was added after the predeclared Paper-65 run, it should be read as exploratory. Its role is to ask whether an independently supplied baryonic concentration axis is related to, independent from, complementary to, or redundant with  $D_{\text{struct}}$ . It is not a confirmatory test of a predeclared  $Q_{\text{bar}}$  hypothesis.

The join is exact:

$$N_{\text{matched}} = 175/175.$$

Across the matched sample,

$$\min Q_{\text{bar}} = 0.2205, \quad \text{median}(Q_{\text{bar}}) = 0.7689, \quad \max Q_{\text{bar}} = 2.5158.$$

| <b>Target</b>                  | $\rho_S(Q_{\text{bar}}, X)$ | <b>Permutation <math>p</math></b> |
|--------------------------------|-----------------------------|-----------------------------------|
| $V_{\text{obs,max}}$           | 0.6175                      | 0.0002                            |
| median bulge-fraction proxy    | 0.5584                      | 0.0002                            |
| $\langle V \rangle$ diagnostic | -0.5469                     | 0.0002                            |
| NFW-like RMSE proxy            | 0.5459                      | 0.0002                            |
| peak $D_{\text{struct}}$       | -0.4783                     | 0.0002                            |
| mean $D_{\text{struct}}$       | -0.4345                     | 0.0002                            |
| NFW-like $\chi^2$ proxy        | 0.3943                      | 0.0002                            |
| median gas-fraction proxy      | -0.3725                     | 0.0002                            |
| $\rho(D_{\text{struct}}, r)$   | 0.3674                      | 0.0002                            |
| mean residual fraction         | -0.3346                     | 0.0002                            |

Table 4: Strongest rank associations of the external  $Q_{\text{bar}}$  input with Paper-65 galaxy-level diagnostics. The permutation  $p$ -values are two-sided and diagnostic only; they do not imply causality.



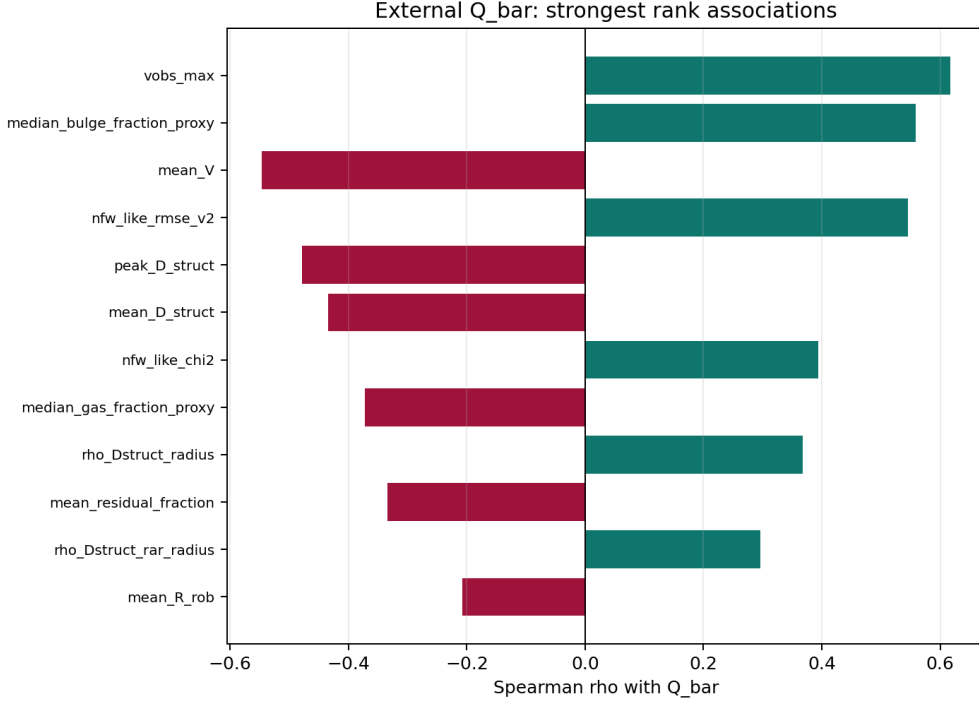


Figure 5: Strongest rank associations of the external  $Q_{\text{bar}}$  input. The strongest positive associations are with  $V_{\text{obs,max}}$ , bulge proxy structure, and NFW-like mismatch proxies. The strongest negative associations include mean and peak  $D_{\text{struct}}$ .

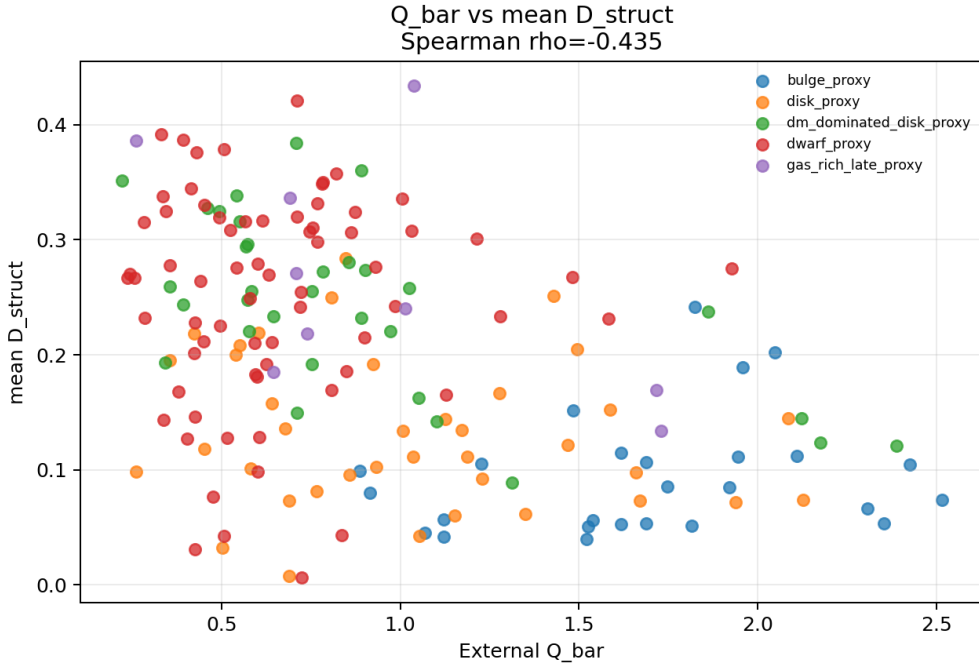


Figure 6: External  $Q_{\text{bar}}$  versus mean  $D_{\text{struct}}$ , coloured by the Paper-65 baryonic morphology proxy. The anti-correlation is visible but is also entangled with galaxy population: high- $Q_{\text{bar}}$  systems are more often bulge/disk-like, while many low- $Q_{\text{bar}}$  systems are dwarf-like or dark-matter-dominated disk proxies.

To test whether the  $Q_{\text{bar}}\text{-}D_{\text{struct}}$  association is only a trivial morphology or mass proxy, we also compute partial rank checks using simple controls. The controls include  $V_{\text{obs,max}}$ , gas/bulge proxies, and a direct mass-size proxy pair derived from the SPARC rotation rows:

$$M_{\text{bar,proxy}}(< R_{\text{max}}) \simeq \frac{V_{\text{bar}}(R_{\text{max}})^2 R_{\text{max}}}{G}, \quad R_{\text{max}} = \max(r).$$

This mass term is a baryonic dynamical mass-scale proxy, not a full photometric stellar-mass estimate. The association is attenuated but not eliminated:

| Target                   | Controls                                  | Partial rank corr. |
|--------------------------|-------------------------------------------|--------------------|
| mean $D_{\text{struct}}$ | $V_{\text{obs,max}}$                      | −0.2148            |
| mean $D_{\text{struct}}$ | gas + bulge proxies                       | −0.2185            |
| mean $D_{\text{struct}}$ | $V_{\text{obs,max}}$ + gas + bulge        | −0.2761            |
| peak $D_{\text{struct}}$ | $V_{\text{obs,max}}$                      | −0.3008            |
| peak $D_{\text{struct}}$ | gas + bulge proxies                       | −0.2883            |
| peak $D_{\text{struct}}$ | $V_{\text{obs,max}}$ + gas + bulge        | −0.3625            |
| mean $D_{\text{struct}}$ | $M_{\text{bar,proxy}}$ + $R_{\text{max}}$ | −0.2593            |
| peak $D_{\text{struct}}$ | $M_{\text{bar,proxy}}$ + $R_{\text{max}}$ | −0.3593            |
| outer residual fraction  | $V_{\text{obs,max}}$ + gas + bulge        | −0.0329            |
| NFW-like $\chi^2$ proxy  | $V_{\text{obs,max}}$ + gas + bulge        | 0.0512             |
| NFW-like $\chi^2$ proxy  | $M_{\text{bar,proxy}}$ + $R_{\text{max}}$ | 0.1482             |
| RAR sigma residual       | $M_{\text{bar,proxy}}$ + $R_{\text{max}}$ | 0.1141             |

Table 5: Partial rank checks for the external  $Q_{\text{bar}}$  input. These are not full causal regressions; they are diagnostic controls against the simplest available mass and baryonic-composition proxies.

As a second independence check, we compare simple five-fold cross-validated linear predictors for the NFW-like and RAR mismatch targets. The comparison uses exactly the three requested feature sets:  $D_{\text{struct}}$ -only,  $Q_{\text{bar}}$ -only, and  $D_{\text{struct}} + Q_{\text{bar}}$ .

| Target             | Model                                | CV $R^2$      | CV $\rho_s$   |
|--------------------|--------------------------------------|---------------|---------------|
| NFW-like $\chi^2$  | $D_{\text{struct}}$ only             | 0.0708        | 0.6211        |
| NFW-like $\chi^2$  | $Q_{\text{bar}}$ only                | 0.1962        | 0.3890        |
| NFW-like $\chi^2$  | $D_{\text{struct}} + Q_{\text{bar}}$ | 0.2086        | 0.5203        |
| NFW-like RMSE      | $D_{\text{struct}}$ only             | 0.2231        | 0.6067        |
| NFW-like RMSE      | $Q_{\text{bar}}$ only                | 0.2040        | 0.5328        |
| NFW-like RMSE      | $D_{\text{struct}} + Q_{\text{bar}}$ | <b>0.2987</b> | <b>0.6654</b> |
| RAR sigma residual | $D_{\text{struct}}$ only             | 0.0607        | 0.2956        |
| RAR sigma residual | $Q_{\text{bar}}$ only                | 0.0148        | 0.1410        |
| RAR sigma residual | $D_{\text{struct}} + Q_{\text{bar}}$ | 0.0587        | 0.2854        |

Table 6: Independence-style prediction check requested for the  $Q_{\text{bar}}$  comparison. The clearest complementarity appears in the NFW-like RMSE channel, where  $D_{\text{struct}} + Q_{\text{bar}}$  improves over both single-feature predictors. The RAR channel does not show such improvement.

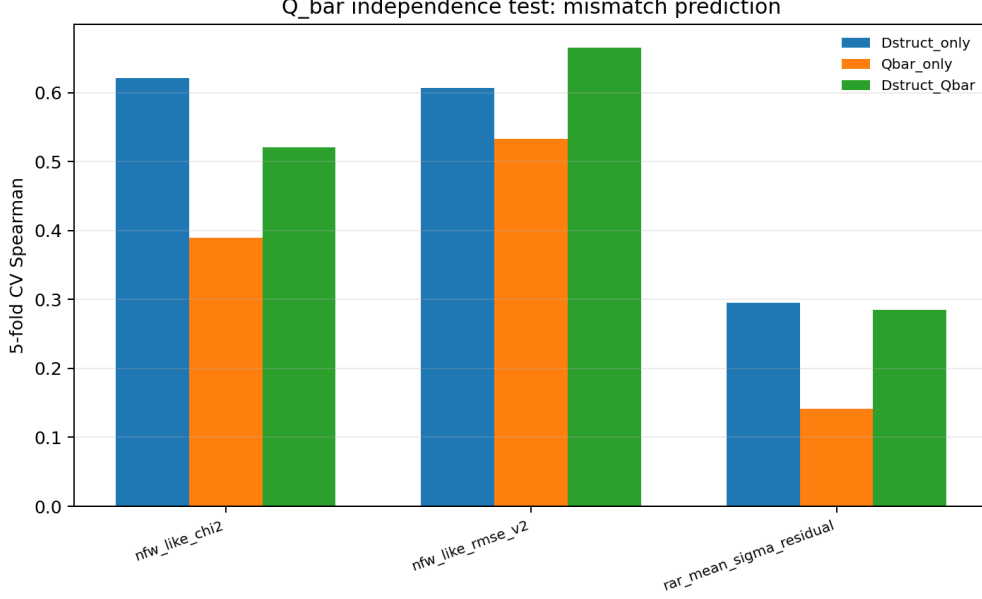


Figure 7: Five-fold cross-validated mismatch prediction using  $D_{\text{struct}}$  only,  $Q_{\text{bar}}$  only, and both together. The result supports a channel-dependent interpretation: partial complementarity in the NFW-like RMSE channel, but no robust added value in the RAR channel.

Quartile summaries show the same pattern. The highest  $Q_{\text{bar}}$  quartile has substantially lower mean  $D_{\text{struct}}$  than the lowest quartile:

$$\langle D_{\text{struct}} \rangle_{Q1} = 0.2429, \quad \langle D_{\text{struct}} \rangle_{Q4} = 0.1283.$$

At the same time, the highest  $Q_{\text{bar}}$  quartile has much larger NFW-like  $\chi^2$ -proxy values. Thus  $Q_{\text{bar}}$  appears to trace a different baryonic concentration and galaxy-population axis rather than simply duplicating  $D_{\text{struct}}$ .

The careful interpretation is therefore:

*$Q_{\text{bar}}$  is not redundant with  $D_{\text{struct}}$ . It is strongly linked to rotation-speed and baryonic morphology proxies, while retaining a moderate anti-correlation with structural residual diagnostics after simple mass-size controls. Its independent predictive contribution is channel-dependent: useful in the NFW-like RMSE channel, weak in the RAR channel. It should be treated as a complementary external baryonic concentration proxy, not as a replacement for  $D_{\text{struct}}$ .*

## 7 Interpretation

Paper 65 strengthens Paper 61 in one important respect:

$$D_{\text{struct}}(r)$$

has radius-resolved information in the NFW-like channel that survives a within-galaxy shuffled-radius null. This is stronger than the Paper-61 shuffled-galaxy result because it preserves each galaxy's structural-score distribution and destroys only the radial assignment.

The result also narrows the claim. The RAR channel does not show the same radius-resolved signal. This is an important negative result. It suggests that the structural residual diagnostic is channel-selective. One possible reading is that the RAR/MOND-like comparison already absorbs a large part of the baryonic radial structure to which  $D_{\text{struct}}$  is sensitive. In that reading,  $D_{\text{struct}}$

and the RAR residual are partly redundant in the baryon-dominated radial information they encode, so little independent rank signal remains after the RAR mapping. Another reading is methodological: the present RAR implementation is a simple sigma-residual channel, not a full fit to the observed RAR with intrinsic scatter, galaxy-specific uncertainties, and covariance structure. A third reading is diagnostic:  $D_{\text{struct}}$  may be tuned to residual-shape support and radial compatibility rather than to acceleration-relation deviations. The current data do not decide between these readings, and the failed RAR channel should therefore be treated as a real constraint, not as a nuisance result. Morphology-proxy splits reveal that the diagnostic behaves differently across galaxy populations. The strongest average structural residual appears in dwarf-like and dark-matter-dominated disk proxies, while the strongest NFW-like radial alignment appears in bulge and disk proxies. Because these classes are derived proxies rather than official morphologies, this population result should be used only to guide a future external morphology test.

Thus the correct interpretation is:

*SPARC structural residuals contain non-random radial information, but the signal is channel- and population-dependent.*

This is not a dark-matter solution. It is a robustness result for a structural diagnostic.

The external  $Q_{\text{bar}}$  comparison adds one further constraint. If the  $Q_{\text{bar}}$  signal were perfectly redundant with  $D_{\text{struct}}$ , it would not add a meaningful independent axis. If it were completely unrelated, it would behave as a null external variable. The observed result lies between those extremes. It is strongly tied to simple galaxy-population proxies, especially maximum rotation speed and bulge fraction, but it still carries a moderate relationship to  $D_{\text{struct}}$ . This makes it useful as an external cross-check and possible covariate in a future multivariate SPARC analysis.

## 8 Required Confirmatory Tests

The present paper narrows the hypothesis space but does not close it. A confirmatory SPARC analysis should be predeclared before rerunning the pipeline and should include:

1. **Published halo fits.** Join the Li-et-al. machine-readable catalogue and replace the internal NFW-like proxy with published multi-halo fit residuals or reduced- $\chi^2$  values.
2. **Official morphology.** Replace the operational proxy classes with an external morphology catalogue or, at minimum, report both official and proxy splits side by side.
3. **Predeclared  $Q_{\text{bar}}$ .** Treat  $Q_{\text{bar}}$  as a predeclared covariate in a multivariate analysis rather than as a post-hoc exploratory check.
4. **Channel asymmetry.** Explicitly test why the NFW-like proxy channel survives the radial null while the RAR/MOND-like channel does not, including whether the RAR residual is redundant with baryonic radial structure, too coarse in the present implementation, or probing a genuinely different diagnostic axis.
5.  **$D_{\text{struct}}$  transparency.** Include a compact restatement of the Paper-61  $D_{\text{struct}}$  construction so that readers can see which ingredients drive the signal.

Only after these steps would it be reasonable to move from “non-random structural diagnostic signal” to a stronger astrophysical model-comparison claim.

## 9 Limitations

- The published Li-et-al. halo-catalogue cross-check is implemented but not executed in this run because no local catalogue table is present.

- Morphology splits are proxy classes, not official Hubble types; they are the weakest component of the population-level analysis.
- Fixed stellar mass-to-light ratios from Paper 61 are retained.
- The NFW-like channel is a lightweight shape proxy, not a professional halo fit.
- The positive NFW-like radius-null result is hypothesis-generating until tested against published halo-fit catalogues.
- The RAR channel fails the shuffled-radius null in this implementation, and the paper does not yet decide whether this reflects redundancy with the RAR mapping, coarseness of the residual channel, or genuine diagnostic channel-selectivity.
- The  $Q_{\text{bar}}$  table is an external derived input and is used only for a neutral post-hoc cross-check in this version.
- The  $Q_{\text{bar}}$  partial-rank checks use simple proxy controls, not a full hierarchical or causal galaxy-population model.
- The present paper restates the operational  $D_{\text{struct}}$  definition but does not fully rederive or theoretically justify it; it uses the Paper-61 diagnostic definition as an input.
- No lensing, CMB, abundance-matching, or full Bayesian halo inference is performed.

## 10 Reproducibility

Repository:

<https://github.com/Chris4081/structural-selection-principle>

Experiment folder:

`experiments/sparc_ii_paper65/`

Run:

```
cd experiments/sparc_ii_paper65
python3 sparc_ii_paper65.py
```

Optional catalogue run:

```
python3 sparc_ii_paper65.py --halo-catalog path/to/li_halo_catalog.csv
```

External  $Q_{\text{bar}}$  comparison:

```
python3 sparc_qbar_external_comparison.py
```

Outputs include:

- `paper65_radius_shuffle_nulls.csv`,
- `paper65_radius_null_summary.json`,
- `paper65_galaxywise_radial_correlations.csv`,
- `paper65_morphology_proxy_splits.csv`,
- `paper65_galaxy_summary_with_proxy_morphology.csv`,
- `paper65_summary.json`,

- four PNG figures,
- `outputs_qbar_external/qbar_correlations.csv`,
- `outputs_qbar_external/qbar_partial_rank_checks.csv`,
- `outputs_qbar_external/qbar_independence_model_comparison.csv`,
- `outputs_qbar_external/qbar_summary.json`,
- five external- $Q_{\text{bar}}$  PNG figures.

**Data attribution and license note.** SPARC data are attributed to Lelli, McGaugh, and Schombert [1]. The local SPARC data used in the repository were prepared from the Zenodo-hosted SPARC rotation-curve record, DOI [10.5281/zenodo.16284118](https://doi.org/10.5281/zenodo.16284118), listed under CC-BY-4.0. Derived CSV, JSON, and PNG files are reproducibility artifacts and must be distinguished from the original SPARC measurements. If a published halo catalogue such as Li et al. [6] is supplied locally, cite the original catalogue source and respect its distribution terms. The external  $Q_{\text{bar}}$  table was supplied by Ali Alhawarat as a derived HGD–GSR comparison descriptor. It is not an original SPARC measurement and not a MAAT-derived quantity. Unless broader redistribution terms are separately supplied by the HGD–GSR author, the table should be treated as a neutral comparison input for this pilot; discussion or reuse should cite or acknowledge Ali Alhawarat and HGD–GSR. No endorsement by SPARC, Li et al., Zenodo, Vizier, CDS, HGD–GSR, Ali Alhawarat, or any original data provider is implied.

## 11 Conclusion

Paper 65 performs the predeclared SPARC-II robustness step. The shuffled-radius null shows that  $D_{\text{struct}}$  retains strong radius-resolved alignment with the NFW-like residual channel. The same is not true for the RAR channel. Morphology-proxy splits show that the signal is population-dependent rather than universal. The external  $Q_{\text{bar}}$  cross-check further shows that a baryonic concentration proxy is strongly population-linked and moderately anti-correlated with  $D_{\text{struct}}$ , even after simple proxy controls.

The result is therefore useful but bounded: SPARC structural residual diagnostics now pass a stronger radial-null test in one channel, and  $Q_{\text{bar}}$  provides a complementary external covariate, but the analysis is not yet a published-halo-catalogue result. The next required step is to supply the Li-et-al. machine-readable halo table, repeat the analysis with official morphology information, include  $Q_{\text{bar}}$  in a multivariate design, and test whether the radius-resolved signal survives against professional multi-halo fits. Until then, the proper conclusion is conservative:

*Paper 65 identifies a channel-selective, non-random SPARC diagnostic signal and a useful external covariate, but it does not yet establish a dark-matter model comparison or a physical explanation of galaxy rotation curves.*

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